

5.6.14. Let $d_0, d_1, d_2, \dots$ be defined by the formula $d_n = 3^n - 2^n$ for every integer $n \geq 0$. Show that this sequence satisfies the following recurrence relation for every integer $k \geq 2$:

$$d_k = 5d_{k-1} - 6d_{k-2}$$

Proof:

To obtain the k^{th} , $(k-1)^{st}$ and $(k-2)^{st}$, we can substitute k , $(k-1)$ and $(k-2)$ in place of n in the explicit formula for $d_0, d_1, d_2, \dots$

$$d_n = 3^n - 2^n$$

$$d_k = 3^k - 2^k$$

$$d_{k-1} = 3^{k-1} - 2^{k-1}$$

$$d_{k-2} = 3^{k-2} - 2^{k-2}$$

$d_k = 5d_{k-1} - 6d_{k-2}$ can be written as:

$$3^k - 2^k = 5(3^{k-1} - 2^{k-1}) - 6(3^{k-2} - 2^{k-2})$$

$$5(3^{k-1} - 2^{k-1}) = 5(3^{k-1}) - 5(2^{k-1})$$

$$6(3^{k-2} - 2^{k-2}) = 6(3^{k-2}) - 6(2^{k-2})$$

$$5(3^{k-1}) - 5(2^{k-1}) - 6(3^{k-2}) + 6(2^{k-2})$$

$$2^{k-1} = 2(2^{k-2})$$

$$3^{k-1} = 3(3^{k-2})$$

$$5(3(3^{k-2})) - 5(2(2^{k-2})) - 6(3^{k-2}) + 6(2^{k-2})$$

$$15(3^{k-2}) - 10(2^{k-2}) - 6(3^{k-2}) + 6(2^{k-2}) =$$

$$9(3^{k-2}) - 4(2^{k-2})$$

$$9(3^{k-2}) - 4(2^{k-2})$$

$$9 = 3^2$$

$$4 = 2^2$$

$$9(3^{k-2}) - 4(2^{k-2}) = 3^2(3^{k-2}) - 2^2(2^{k-2}) = 3^k - 2^k$$

$$d_k = 3^k - 2^k$$

Hence, $d_k = 5d_{k-1} - 6d_{k-2} = 3^k - 2^k$, which is what was to be show. □

Solution: $d_k = 5d_{k-1} - 6d_{k-2} = 3^k - 2^k$ for every integer $k \geq 2$.